# Conservation laws as Lie-algebra membership: a unified classroom view of state-dependent matrix flows

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**Abstract**

Conservation laws are usually taught one at a time: the length of a rotating vector, the energy of an oscillator, the total amount of a diffusing substance, the spacetime interval under a Lorentz boost. Each comes with its own proof. We show that when the equations of motion are written as a matrix multiplying the state, all four follow from one short calculation. A quantity is conserved when the matrix satisfies a simple algebraic condition, the same condition that defines the symmetry group of that quantity. The idea is known, and we cite its sources; our aim is to make it teachable. We use it to illustrate three points that students often miss. First, the same motion can be described by different matrices, and each may reveal a different conserved quantity. Second, conserving a quantity is not the same as being Hamiltonian: we give a simple three-dimensional example in which a quantity is conserved exactly, yet every trajectory converges to a single point. Third, on a computer, how the error of a numerical method behaves over long times matters more than its size over short times. The text is aimed at graduate courses in classical mechanics and mathematical methods, and every result is reproduced by a short script.

## 1 Introduction

A rotating vector keeps its length, a frictionless oscillator keeps its energy, a diffusing dye keeps its total amount, and a Lorentz boost keeps the interval $t^2 - x^2$. Each fact is usually proved where it first appears, with tools specific to that course. They are, however, the same fact. Write the state as a vector $s \in \mathbb{R}^n$ and the dynamics as

$$\dot{s} = A(s)\, s, \tag{1}$$

with a matrix $A(s)$ that may depend on the state. For a constant symmetric matrix $G$,

$$\frac{d}{dt}\left(s^{\mathsf{T}} G s\right) = s^{\mathsf{T}}\left(A^{\mathsf{T}} G + G A\right) s, \tag{2}$$

so the quadratic form $s^{\mathsf{T}}Gs$ is conserved whenever

$$A(s)^{\mathsf{T}}G + GA(s) = 0 \quad \text{for all } s. \tag{3}$$

For invertible $G$, condition (3) says that $A(s)$ belongs to the Lie algebra $\mathfrak{so}(G)$ of linear maps that preserve $G$ infinitesimally. A conservation law becomes a statement about which algebra the generator belongs to.

Nothing here is new mathematics, and we want to be explicit about it. The factorization $f(s) = A(s)s$ and its non-uniqueness are the starting point of the state-dependent Riccati equation (SDRE) method in control theory, where $A(s)$ is called a state-dependent coefficient [1, 2]. That any flow with a first integral $I$ can be written as $S(s)\nabla I$ with $S$ skew is the unified approach of McLachlan, Quispel and Robidoux [3]. Brackets that are skew but fail the Jacobi identity are the almost-Poisson brackets of nonholonomic mechanics [4, 5, 6, 7]. Integrators that respect the algebra of $A(s)$ are the Lie-group methods of Iserles *et al.* [8], and the long-time behaviour of symplectic integrators is the subject of geometric numerical integration [9].

What these sources do not do, because it is not their purpose, is put the elementary cases side by side so that a student sees one mechanism behind them. That is the aim of this paper. Section 2 shows that every smooth flow fixing the origin has the form (1), and that the freedom in choosing $A$ is real: different choices certify different invariants. Section 3 collects the familiar conservation laws into one table. Section 4 shows that condition (3) is the same as an almost-Poisson structure and gives a three-dimensional example, solvable by hand, where the Jacobi identity fails and the consequences are visible. Section 5 shows what survives discretization. The material suits a graduate course in classical mechanics or mathematical methods of physics, and parts of Sections 2 and 3 work in an advanced undergraduate course on linear algebra and differential equations.

All numbers and both figures in this paper are produced by one script, `tools/journal_math.py`, in an open code base built around (1) [16]. It asserts every identity below to machine precision and integrates every example.

## 2 The form and its freedom

Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable with $f(0) = 0$. Then

$$f(s) = \int_0^1 \frac{d}{d\tau} f(\tau s)\, d\tau = \left[\int_0^1 Df(\tau s)\, d\tau\right] s \equiv A_0(s)\, s, \tag{4}$$

so every such vector field can be written as (1). Fixed points elsewhere are moved to the origin by a shift. The form is therefore no restriction on the dynamics; what it adds is a matrix to inspect.

That matrix is not unique. If $B(s)\, s = 0$ for every $s$, then $A$ and $A + B$ generate the same vector field. In three dimensions, for example, $B(s) = \widehat{u}(s)\, \widehat{s}$ works for any vector field $u$, where $\widehat{v}$ denotes the matrix with $\widehat{v}x = v \times x$. Since (3) is a property

Table 1: Conservation laws as membership. The last column is the largest entry of the certificate for a random member of the class, as computed by the accompanying script; zero means the identity holds exactly in floating point.

| class of $A$ | $G$ | conserved quantity | residual |
|---|---|---|---|
| antisymmetric, $\mathfrak{so}(n)$ | 1 | $\lvert s\rvert^2$ | 0 |
| $A = JM$, $M = M^{\mathsf{T}}$ | $M$ | energy $\frac{1}{2}s^{\mathsf{T}}Ms$ | $2 \times 10^{-14}$ |
| same | $J$ | symplectic form | 0 |
| realified anti-Hermitian, $\mathfrak{u}(n)$ | 1 | $\langle\psi\lvert\psi\rangle$ | 0 |
| same | $J_c$ | symplectic form | 0 |
| Lorentz, $\mathfrak{so}(3,1)$ | $\eta$ | $t^2 - \lvert\boldsymbol{x}\rvert^2$ | 0 |
| graph Laplacian | $v = (1, \ldots, 1)$ | total mass $v^{\mathsf{T}}s$ | 0 |

of the matrix and not of the vector field, the choice of $A$ decides which invariant one can read off. The certificate is sufficient, never necessary for a given factorization.

The free rigid body makes this concrete. With angular momentum $\boldsymbol{m}$ in the body frame and principal moments $I = \mathrm{diag}(I_1, I_2, I_3)$, Euler's equations read $\dot{\boldsymbol{m}} = \boldsymbol{m} \times I^{-1}\boldsymbol{m}$. Two factorizations suggest themselves:

$$A_1(\boldsymbol{m}) = \widehat{\boldsymbol{m}}\, I^{-1}, \qquad A_2(\boldsymbol{m}) = -\widehat{I^{-1}\boldsymbol{m}}. \tag{5}$$

Both give $A_i(\boldsymbol{m})\boldsymbol{m} = \boldsymbol{m} \times I^{-1}\boldsymbol{m}$. The first satisfies (3) with $G = I^{-1}$, because $A_1^{\mathsf{T}}I^{-1} + I^{-1}A_1 = -I^{-1}\widehat{\boldsymbol{m}}I^{-1} + I^{-1}\widehat{\boldsymbol{m}}I^{-1} = 0$, and so certifies the kinetic energy $E = \frac{1}{2}\boldsymbol{m}^{\mathsf{T}}I^{-1}\boldsymbol{m}$. The second is antisymmetric, so it certifies $\lvert\boldsymbol{m}\rvert^2$ with $G = 1$. Neither certifies the other's invariant: for $I = \mathrm{diag}(1, 2, 3)$ and random states, the residual of $A_1$ in $\mathfrak{so}(1)$ is at least 0.21 and that of $A_2$ in $\mathfrak{so}(I^{-1})$ at least 0.11, while both vector fields agree to $1 \times 10^{-16}$. The one vector field has two conserved quantities, and each factorization exhibits one of them.

The converse also holds, and it is the content of [3]. If a vector field $f$ conserves $Q = \frac{1}{2}s^{\mathsf{T}}Gs$, then $f \cdot \nabla Q = 0$, and wherever $\nabla Q \neq 0$

$$S(s) = \frac{f\,\nabla Q^{\mathsf{T}} - \nabla Q\, f^{\mathsf{T}}}{\lvert\nabla Q\rvert^2} \tag{6}$$

is skew and satisfies $S\nabla Q = f$. Since $\nabla Q = Gs$, the matrix $A = SG$ is a factorization in $\mathfrak{so}(G)$. For the rigid body with $Q = E$ the script finds (6) skew and exact to $1 \times 10^{-16}$. So a quadratic form is conserved if and only if *some* factorization lies in its algebra, away from $\nabla Q = 0$.

# 3 One equation, many invariants

Table 1 collects the familiar cases. Each row is the same equation (3) with a different $G$.

*Norm.* An antisymmetric $A$ rotates $s$ and keeps its length. The Schrödinger equation $i\dot{\psi} = H\psi$ is of this type once the complex vector $\psi = x + iy$ is written as

the real vector $s = (x, y)$: the generator $-iH$ becomes

$$A = \begin{pmatrix} \operatorname{Im} H & \operatorname{Re} H \\ -\operatorname{Re} H & \operatorname{Im} H \end{pmatrix}, \tag{7}$$

which is antisymmetric precisely because $H$ is Hermitian. Unitarity is the norm row of the table.

*Energy.* For a linear Hamiltonian system with $H = \frac{1}{2} s^{\mathsf{T}} M s$ and $s = (q, p)$, Hamilton's equations are $\dot{s} = JMs$ with $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Then $A^{\mathsf{T}}M + MA = MJ^{\mathsf{T}}M + MJM = 0$, so the energy is the $G = M$ row. The same $A$ also satisfies $A^{\mathsf{T}}J + JA = 0$, the statement that $A$ lies in the symplectic algebra $\mathfrak{sp}(2n)$; this is a statement about the phase-space area form, not about a conserved number.

*Quantum mechanics is both.* The realified anti-Hermitian generators preserve both the norm ($G = 1$) and the complex structure $J_c$. They are the intersection $\mathfrak{so}(2n) \cap \mathfrak{sp}(2n) = \mathfrak{u}(n)$, which is the algebraic content of the remark that the Schrödinger equation is a Hamiltonian system [10].

*Interval.* The generators of boosts and rotations satisfy (3) with $\eta = \operatorname{diag}(1, -1, -1, -1)$. A boost along $x$ has $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ on $(t, x)$, which is symmetric, not antisymmetric; it is $\eta$-antisymmetric, and that is all (2) needs.

*Mass.* A graph Laplacian $L$ on a network has vanishing column sums, $v^{\mathsf{T}}L = 0$ with $v = (1, \ldots, 1)$. The linear form $v^{\mathsf{T}}s$, the total mass, is then conserved, by the degree-one version of (2): $\frac{d}{dt}(v^{\mathsf{T}}s) = v^{\mathsf{T}}As$. The same matrix is *not* in $\mathfrak{so}(1)$ (residual 4.0 for an eight-node chain): its symmetric part is negative semidefinite, so $|s|^2$ decays. Diffusion keeps the total and spreads it out, and the table separates it from the conservative rows rather than merely labelling it.

# 4 Conservative is almost-Poisson; Jacobi is the rest

For invertible symmetric $G$ define $\Pi(s) = A(s)\, G^{-1}$ and $H(s) = \frac{1}{2} s^{\mathsf{T}} G s$. Then $\nabla H = Gs$ and

$$\dot{s} = A(s)\, s = \Pi(s)\, \nabla H. \tag{8}$$

**Proposition 1.** *$A(s)^{\mathsf{T}}G + GA(s) = 0$ if and only if $\Pi(s)$ is antisymmetric.*

*Proof.* $\Pi^{\mathsf{T}} = G^{-1}A^{\mathsf{T}}$, so $\Pi^{\mathsf{T}} = -\Pi$ is $G^{-1}A^{\mathsf{T}} = -AG^{-1}$; multiply by $G$ on both sides. □

The script checks both directions with an indefinite $G$: a matrix built as $A = \Pi G$ from a random antisymmetric $\Pi$ has zero residual, and a generic $A$ fails both conditions at order one. An antisymmetric $\Pi(s)$ defines the bracket

$$\{F, K\}(s) = \nabla F^{\mathsf{T}}\, \Pi(s)\, \nabla K, \tag{9}$$

which is bilinear, antisymmetric and obeys the Leibniz rule, and (8) is $\dot{F} = \{F, H\}$. Energy conservation is $\{H, H\} = 0$, nothing more. Such a bracket is called almost-Poisson. It is a Poisson bracket, and the dynamics is Hamiltonian in the full sense,

only if it also satisfies the Jacobi identity, which in coordinates is the vanishing of the Jacobiator

$$J^{ijk} = \Pi^{il}\partial_l\Pi^{jk} + \Pi^{jl}\partial_l\Pi^{ki} + \Pi^{kl}\partial_l\Pi^{ij} \tag{10}$$

(summation over $l$). Jacobi is what delivers the rest of Hamiltonian mechanics: the flow preserves $\Pi$ itself, phase space splits into symplectic leaves, Darboux coordinates exist on each, and Liouville's theorem holds on them [10, 11].

Three cases are immediate. (i) If $A$ is constant, so is $\Pi$, every derivative in (10) vanishes, and every *linear* conservative system is Poisson. (ii) If $\Pi$ is linear in $s$, $\Pi^{ij} = c^{ij}_k s^k$, the Jacobi identity for $\Pi$ is the Jacobi identity for the structure constants $c^{ij}_k$: these are the Lie–Poisson brackets. The rigid body is the example, with $\Pi = \widehat{\boldsymbol{m}}$ from $A_1$ in (5) and the structure constants of $\mathfrak{so}(3)$. The second factorization $A_2$ also gives a Poisson bracket: $\Pi_2 = -\widehat{\nabla E}$ turns Euler's equations into $\dot{\boldsymbol{m}} = \nabla C \times \nabla E$ with $C = \frac{1}{2}|\boldsymbol{m}|^2$, Nambu's form [12]. The script finds the Jacobiator of both exactly zero. (iii) In two dimensions every antisymmetric $\Pi$ satisfies Jacobi, because the Jacobiator is totally antisymmetric in $i, j, k$ and a totally antisymmetric three-index object in two dimensions vanishes. A conservative, non-Hamiltonian example therefore needs $n \geq 3$. This is a trap: reduced two-dimensional nonholonomic models, which otherwise look like the natural examples, cannot exhibit the failure.

## 4.1 A conservative flow that is not Hamiltonian

In three dimensions every antisymmetric matrix can be written $\Pi^{ij} = \epsilon^{ijk} w_k$ for a vector $w(s)$, and $\Pi\nabla H = \nabla H \times w$. A short computation, or the script, gives the one independent component of (10):

$$J^{123} = -\,w \cdot (\nabla \times w). \tag{11}$$

The Jacobi identity is the Frobenius condition for the planes orthogonal to $w$ to be tangent to a family of surfaces [13, 14]. Choose a $w$ whose planes twist:

$$w(s) = (-s_2,\ s_1,\ 1), \qquad \nabla \times w = (0, 0, 2), \qquad J^{123} = -2. \tag{12}$$

With $G = 1$ and $H = \frac{1}{2}|s|^2$, the flow $\dot{s} = s \times w$ reads

$$\dot{s}_1 = s_2 - s_1 s_3, \qquad \dot{s}_2 = -s_1 - s_2 s_3, \qquad \dot{s}_3 = s_1^2 + s_2^2. \tag{13}$$

The generator $A = \Pi$ is antisymmetric, its certificate is exactly zero in floating point, and $|s|^2$ is conserved. The Jacobiator is $-2$ at every point, evaluated by central differences, which are exact here because $\Pi$ is affine in $s$. On the unit sphere the third equation closes on itself, $\dot{s}_3 = 1 - s_3^2$, with the solution

$$s_3(t) = \tanh(t + \operatorname{artanh} s_3(0))\,. \tag{14}$$

Every orbit except the south pole climbs to the north pole (figure 1). Integrated with fourth-order Runge–Kutta at step $10^{-3}$, $s_3$ follows the closed form to $2.5 \times 10^{-14}$ and $|s|^2$ stays at 1 to $2.8 \times 10^{-14}$.

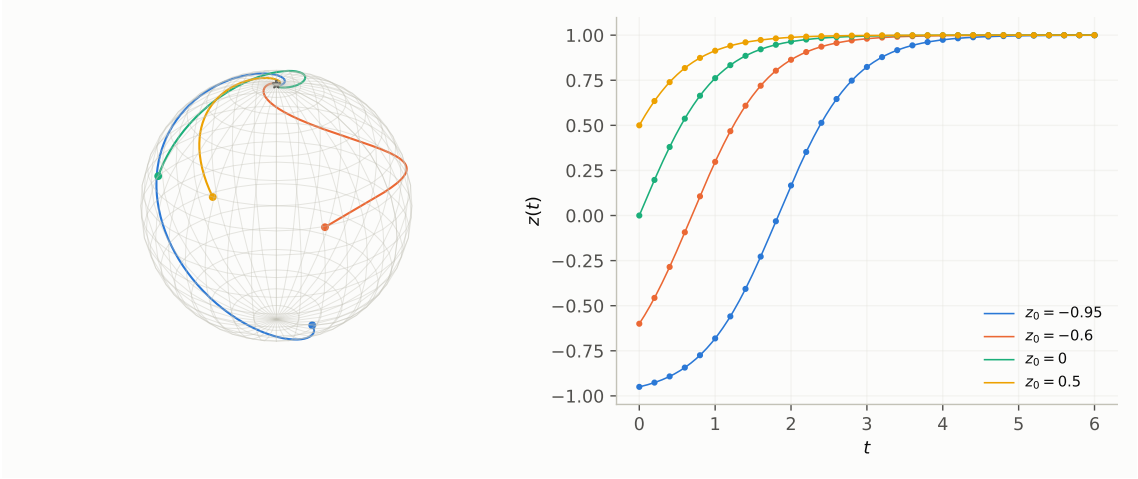



Figure 1: The flow (13). Left: four orbits on the invariant unit sphere (dots mark the starts), all drawn into the north pole (star). Right: $s_3(t)$ from the integration (dots) against $\tanh(t + \operatorname{artanh} s_3(0))$ (lines).

An attracting fixed point is exactly what an area-preserving flow cannot have: a small disc around the pole would be mapped into ever smaller discs inside itself. So the flow (13), restricted to its invariant sphere, preserves no area form there, and it is not the Hamiltonian flow of any symplectic structure on the sphere. Conservation of $|s|^2$, which the certificate guarantees, is not enough. The pole attracts while the energy stays put; what is lost is Liouville's theorem, and the Jacobi identity is precisely the condition that would have restored it. In mechanics such brackets arise from nonholonomic constraints, such as rolling without slipping, where energy is conserved and phase-space volume generally is not [4, 5, 6].

# 5 What survives discretization

A computer replaces (1) by a map $s_{k+1} = \Phi_h(s_k)$, and the question becomes which invariants $\Phi_h$ keeps. Every Runge–Kutta method keeps linear invariants such as the total mass exactly; quadratic invariants are kept exactly only by the symplectic Runge–Kutta methods, such as the implicit midpoint rule [9, section IV.2]. The explicit fourth-order method (RK4) is not among them. A symplectic method such as velocity Verlet does not conserve the energy exactly either, but it conserves a nearby modified energy, so its energy error oscillates in a bounded band for exponentially long times instead of growing [9, chapter IX].

Two numbers are needed to see this, and one is not enough. The *band* is $(\max H - \min H)/|\bar{H}|$ over the run; the *leak* is the difference between the mean of $H$ over the last and the first 5% of the run, divided by $|\bar{H}|$. Take two uncoupled oscillators, $H = \frac{1}{2}(q_1^2 + 2.5\,q_2^2 + p_1^2 + p_2^2)$, a deliberately coarse step $h = 0.15$, and integrate to $t = 8000$:

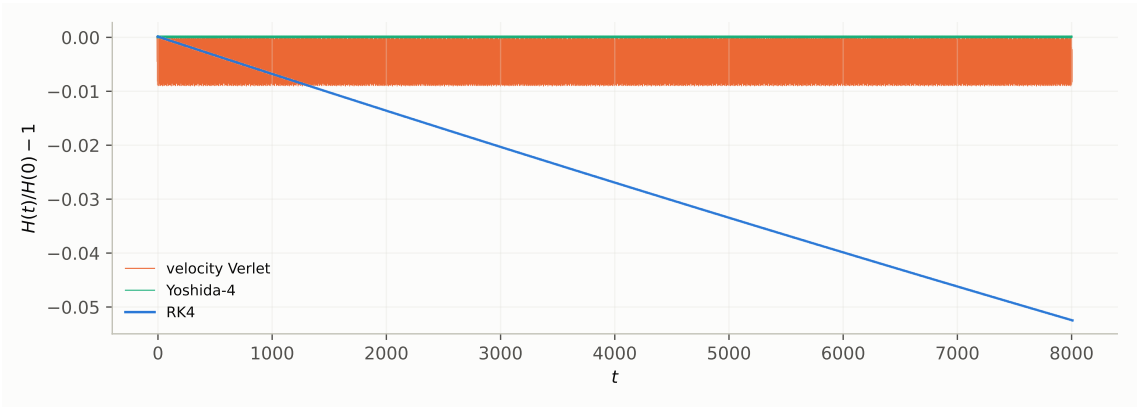



Figure 2: Relative energy error on the run of the table. Verlet fills a fixed band; RK4 leaks linearly and leaves that band for good at $t \approx 1290$. Yoshida's band is too thin to see at this scale.

| method | band | leak |
|---|---|---|
| RK4 | $5.4 \times 10^{-2}$ | $5.1 \times 10^{-2}$ |
| velocity Verlet (order 2) | $8.9 \times 10^{-3}$ | $4.4 \times 10^{-6}$ |
| Yoshida (order 4) [15] | $1.2 \times 10^{-4}$ | $1.2 \times 10^{-7}$ |

The RK4 error is almost entirely leak; Verlet's is almost entirely band (figure 2). At short times this ranks them the wrong way round. Verlet's band reaches $8.9 \times 10^{-3}$ immediately, while RK4's error only exceeds that value at $t \approx 1290$, after some 300 periods. A comparison by maximum error over a run shorter than that would declare RK4 the better integrator. The leak exposes the difference at once, and the conclusion does not depend on the length of the run. We have found this a useful exercise: students who compare methods by a single error number usually pick the wrong one here.

# 6 Suggestions for the classroom

The material lends itself to short exercises, each checkable in a few lines of code. (a) Show that the boost generator is symmetric yet conserves $t^2 - x^2$, and find the $G$ for which a given $2 \times 2$ matrix is a certificate. (b) Find a third factorization of Euler's equations and identify what, if anything, it certifies. (c) Compute (10) for $\Pi^{ij} = \epsilon^{ijk} w_k$ and derive (11); then show that the Jacobiator vanishes identically in two dimensions. (d) Solve (13) on the sphere and show that the pole attracts; find a curve along which the flow strictly increases a function, and explain why that rules out a preserved area form. (e) Reproduce the table of section 5 and find the step $h$ at which the crossover time drops below one period.

# 7 Conclusion

Conservation of a quadratic quantity under $\dot{s} = A(s)s$ is the statement that $A$ belongs to the Lie algebra of the form's isometry group. This single equation organizes the conservation laws of an undergraduate physics curriculum into one table. Because the factorization is not unique, each choice of $A$ certifies its own invariant. Because conservation is only almost-Poisson, it does not by itself make the dynamics Hamiltonian; the Jacobi identity does, and it can fail only from three dimensions up, as a closed-form example shows. And because computers discretize, the long-time quality of an integrator shows up in its leak, not its band. All of this is known; we hope that putting it in one place makes it easier to teach.

# Data availability

The script that reproduces every number and figure, `tools/journal_math.py`, is part of the archived code base [16].

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